# AFFORD-Bench: An Affordance-Grounded Benchmark and Generalization Evaluation for Robot Learning

Author Names Omitted for Anonymous Review. Paper-ID [RSS-2027-0150]

***Abstract*—Imitation learning policies and Vision-Language-Action (VLA) models have demonstrated remarkable empirical success in robotic manipulation. However, recent benchmark analyses, such as LIBERO-PRO and LIBERO-X, have revealed a fundamental crisis in evaluation: high task completion rates frequently collapse under even minor environmental perturbations, exposing policies that rely on rote trajectory memorization rather than genuine environmental comprehension. Crucially, existing benchmarks evaluate distribution shifts predominantly in narrow tabletop settings using superficial appearance or instruction variations, ignoring the physical affordances—contact mechanics, kinematic clearances, dynamic friction, and compliance—that govern physical interaction. In this paper, we introduce AFFORD-Bench, a large-scale, affordance-grounded manipulation benchmark spanning 150 diverse physical tasks across 5 distinct domains: Industrial Precision, Culinary Operations, Domestic Tabletop, Cleanroom Laboratory Sciences, and Long-Horizon Multi-Stage Sequences. Operating in a high-fidelity Franka Emika Panda simulation environment, AFFORD-Bench provides 7,500 expert demonstrations across dual-camera high-resolution RGB ($640 \times 660$ workcell and $300 \times 420$ wrist) alongside aligned 3D metric depth and 7-DoF impedance states. We formalize a systematic combinatorial perturbation manifold spanning Spatial-Kinematic ($\pm\Delta$p, orientation tilts, insertion tolerances), Physical-Dynamic (Coulomb friction $\mu \in [0.1, 1.2]$, mass scaling $0.5\times$–$3.0\times$), and Visual-Photometric axes. We detail comprehensive training and evaluation protocols, establishing standardized metrics including the Physical Robustness Index (PRI). Our empirical evaluations across leading policy representations—including Diffusion Policy, ACT, OpenVLA, and recent compact and subtask-aware VLAs such as SmolVLA and AtomVLA—demonstrate that state-of-the-art architectures degrade non-uniformly along affordance boundaries, confirming that physical affordance grounding—rather than appearance scaling alone—is necessary for generalizable manipulation.***

## I. Introduction

The field of robot learning is undergoing a profound paradigm shift driven by the emergence of generalist Vision-Language-Action (VLA) models [9, 12, 13], compact edge architectures such as SmolVLA [3], subtask-aware post-training frameworks like AtomVLA [19], and expressive visuomotor policies [4, 20]. By unifying multi-modal perception with end-to-end motor control, these architectures promise to endow robotic manipulators with cross-task generalization and zero-shot adaptability.

However, measuring whether these models exhibit true physical generalizability remains an acute open challenge. Standardized evaluation suites such as LIBERO [10] have become the de facto testing ground for policy comparison. Yet, rigorous post-hoc investigations by Zhou et al. [21] (LIBERO-PRO) and Wang et al. [17] (LIBERO-X) revealed that the seemingly high success rates reported on standard benchmarks are severely inflated. When subjected to minor spatial offsets or altered visual conditions, policies that previously achieved over 90% success suffer catastrophic degradation down to near 0%. As established in both analyses, current models overfit to rote trajectory memorization and nominal scene layouts from demonstration data, lacking any grounded understanding of the task goals.

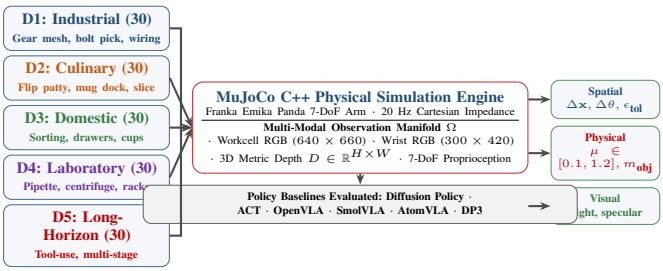



Fig. 1. **The AFFORD-Bench Architecture.** Spanning 150 tasks across 5 distinct domains, benchmarking out-of-distribution generalization across Spatial-Kinematic, Physical-Dynamic, and Visual-Photometric shifts.

While LIBERO-PRO and LIBERO-X took vital steps by introducing independent and hierarchical perturbations, existing benchmarks remain constrained by a critical limitation: **the absence of physical affordance grounding**. In ecological psychology and robotics [6], an *affordance* defines the actionable properties between an agent and its physical environment. Real-world robotic manipulation is not merely a geometric or visual alignment problem; it is fundamentally governed by contact mechanics, surface friction, kinematic compliance, mass distributions, and tight clearance tolerances. In existing suites, tasks are largely limited to unconstrained tabletop pick-and-place operations in toy domestic scenes, where collisions and contact physics are trivialized. Consequently, they fail to evaluate whether a policy can adapt to variations in physical contact constraints (e.g., insertion resistance, grasping friction, pipette liquid aspiration, or high-tolerance gear meshing).

To bridge this foundational gap, this paper introduces **AFFORD-Bench**, an affordance-grounded benchmark designed to expose the boundary between visual memorization and physical generalizability. AFFORD-Bench advances the state of robot learning along four primary axes:

1) **Scale and Domain Diversity**: We formulate **150 distinct physical manipulation tasks** distributed across **5 specialized domains**: (i) *Industrial Precision & Manufacturing*, (ii) *Culinary & Commercial Kitchen*, (iii) *Domestic Tabletop*, (iv) *Cleanroom Laboratory Sciences*, and (v) *Long-Horizon Multi-Stage Sequences*.
2) **Affordance-Grounded Perturbation Space**: Unlike prior benchmarks that alter visual textures or object coordinates in isolation, we formalize a structured perturbation manifold across three orthogonal physical axes: *Spatial-Kinematic* (clearance tolerances $\pm 2$ mm to $\pm 10$ mm, orientation tilts), *Physical-Dynamic* (Coulomb friction $\mu \in [0.1, 1.2]$, object masses $0.5\times$ to $3.0\times$, contact damping), and *Visual-Photometric* (lighting angle, specularity, distractor clutter).
3) **Multi-Modal Metric Perception**: All environments feature a 7-DoF Franka Emika Panda arm equipped with synchronized dual-camera visual streams (workcell $640 \times 660$ and wrist $300 \times 420$), true 3D metric depth, and 7-DoF joint/Cartesian impedance proprioception.
4) **High-Throughput, Zero-Cost Reproducibility**: Using native C++ MuJoCo physics [16], we synthesize 7,500 expert demonstrations (50 per task, $> 1.5 \times 10^6$ timesteps) via deterministic algorithmic oracles running at 40–50 FPS with zero external API costs. Datasets are directly packaged in standardized Robomimic/HDF5 format compatible with Diffusion Policy, ACT, OpenVLA, SmolVLA, AtomVLA, and Octo.

## II. Related Work

### A. Robotic Manipulation Benchmarks

Benchmarking in robotic manipulation has evolved from single-domain suites to diverse simulation platforms. RLBench [8] and ManiSkill2 [7] provided large sets of simulation tasks for reinforcement learning and demonstration-guided policy learning. Robomimic [11] established standardized offline imitation learning pipelines and empirical evaluation protocols across varied human demonstration qualities. LIBERO [10] introduced a 130-task suite structured around knowledge transfer in lifelong learning, evaluating sequential task acquisition. However, all these suites focus primarily on nominal execution without systematic physical out-of-distribution (OOD) testing.

### B. OOD Robustness and Modern VLA Architectures

With the proliferation of large-scale pretrained models [13, 2, 1, 9, 12], assessing policy robustness against distribution shifts has become critical. Zhou et al. [21] established LIBERO-PRO, illustrating that state-of-the-art policies achieve high success under nominal LIBERO conditions but collapse when subject to perturbed objects, initial states, or paraphrased instructions. Building on this, Wang et al. [17] introduced LIBERO-X, demonstrating that prior benchmarks fail to evaluate compounded distribution shifts across hierarchical difficulty levels.

Concurrently, the community has pushed toward efficient edge-deployable VLAs like **SmolVLA** [3], which utilizes a 450M parameter vision-language backbone with a flow-matching action expert to drastically reduce inference latency, and **AtomVLA** [19], which introduces atomic subtask decomposition paired with a predictive world model to mitigate error accumulation in long-horizon tasks. However, existing benchmarks evaluate these modern architectures primarily within standard domestic tabletop scenes, failing to probe contact dynamics, surface friction, or precision assembly.

### C. Affordance and Visuomotor Policy Learning

Affordance theory [6] posits that agents perceive objects in terms of potential physical interactions. In robotics, affordance-guided manipulation has been explored via contact-rich grasping [15], spatial affordance maps [14], and explicit 3D representations such as 3D Diffusion Policy (DP3) [18]. While Diffusion Policy [4] and Action Chunking with Transformers (ACT) [20] achieve high tracking precision on low-cost hardware [5], their ability to generalize across physical affordance shifts remains uncharacterized at scale. AFFORD-Bench provides the first systematic benchmark unifying multi-domain manipulation with explicit affordance-space perturbation manifolds.

## III. The AFFORD-Bench Benchmark

### A. Task Formulation as Affordance-Constrained MDPs

We formalize each manipulation task as an Affordance-Constrained Markov Decision Process defined by the tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \Omega, \mathcal{C}_{\text{aff}} \rangle$, where $\mathcal{S}$ is the continuous state space of the robot and environment, $\mathcal{A} \subset \mathbb{R}^4$ is the action space representing 3D Cartesian end-effector delta velocity $\Delta \mathbf{x} \in \mathbb{R}^3$ and a continuous gripper clamping command $a_{\text{grip}} \in [-1, 1]$, $\mathcal{T}(s' \mid s, a)$ is the transition probability governed by MuJoCo multi-body contact dynamics, and $\mathcal{R}(s, a) \in \{0, 1\}$ is a sparse task completion indicator.

Crucially, $\mathcal{C}_{\text{aff}} \subset \mathcal{S} \times \mathcal{A}$ denotes the *affordance constraint set*. For a task to be successfully executed, the state trajectory must satisfy both geometric tolerances and dynamic contact stability:

$$\mathcal{C}_{\text{aff}} = \Big\{ (s, a) \;\Big|\; \|\mathbf{p}_{\text{obj}} - \mathbf{p}_{\text{goal}}\|_{\mathbf{W}} \le \epsilon_{\text{tol}}, \; F_{\text{contact}} \le F_{\max}, \; \frac{F_{\text{friction}}}{F_{\text{normal}}} \le \mu \Big\}, \tag{1}$$

where $\mathbf{W} = \text{diag}(w_x, w_y, w_z)$ weights directional tolerance, $\epsilon_{\text{tol}}$ represents the insertion or placement threshold, and $\mu$ is the Coulomb friction coefficient.

### B. Domain Architecture

As illustrated in Table I and visualized in Figure 2, AFFORD-Bench is partitioned into five specialized manipulation domains, each comprising exactly 30 distinct tasks:

- **D1: Industrial Precision & Manufacturing (Tasks 001–030)**: Tasks focus on assembly line operations requiring sub-centimeter accuracy, including gear meshing onto splined shafts, flange bolt pickup and fixture insertion, cable harness threading, and toggle clamp engagement.
- **D2: Culinary & Commercial Kitchen (Tasks 031–060)**: Operations simulate automated commercial kitchen preparation,

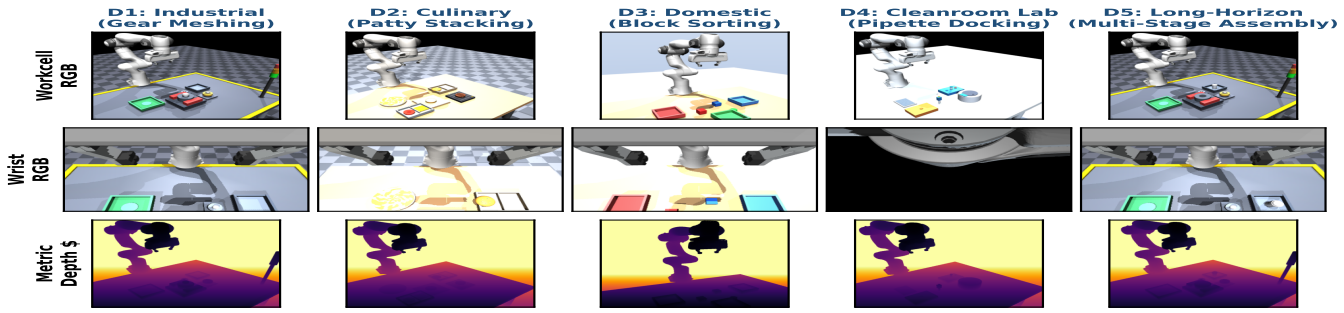



Fig. 2. **Multi-Domain Qualitative Visualizations in AFFORD-Bench.** Synchronized multi-modal observations across representative tasks from the five benchmark domains: (Row 1) Third-person workcell visual stream ($640 \times 660$ RGB); (Row 2) Wrist-mounted eye-in-hand visual stream ($300 \times 420$ RGB); (Row 3) Aligned 3D metric depth maps ($D_{3D}$, colored by physical meter distance). Spans industrial precision assembly, commercial culinary preparation, domestic block sorting, cleanroom pharmaceutical sciences, and multi-stage long-horizon sequences.

TABLE I

COMPREHENSIVE TAXONOMY OF THE AFFORD-BENCH SUITE. SPANNING 150 FORMALLY DEFINED PHYSICAL TASKS ACROSS 5 DISTINCT DOMAINS, FEATURING DIVERSE KINEMATIC CONSTRAINTS, CONTACT DYNAMICS, AND CLEARANCE TOLERANCES.

| Domain | Task IDs | Count | Representative Manipulation Tasks | Clearance ($\epsilon$) | Dominant Contact Physics |
|---|---|---|---|---|---|
| **D1: Industrial Precision** | 001–030 | 30 | Gear meshing, bolt insertion, cable routing, fixture latching | $\pm$2.0–5.0 mm | High stiffness, rigid peg-in-hole friction |
| **D2: Culinary Operations** | 031–060 | 30 | Patty flipping, espresso mug docking, slicing alignment, pouring | $\pm$5.0–12.0 mm | Deformable contact, non-planar slipping |
| **D3: Domestic Tabletop** | 061–090 | 30 | Color block sorting, drawer opening, stack packing, cup nesting | $\pm$8.0–20.0 mm | Multi-body clutter, dynamic toppling |
| **D4: Laboratory Sciences** | 091–120 | 30 | Pipette tip pickup, centrifuge vial docking, reagent mixing | $\pm$1.5–4.0 mm | Precision alignment, fragile glass contact |
| **D5: Long-Horizon Sequences** | 121–150 | 30 | Multi-stage assembly, tool-assisted latching, sequential routing | $\pm$3.0–10.0 mm | Cumulative error accumulation, subgoal chaining |

including precision spatula patty flips, espresso portafilter and mug positioning, ingredient transfer, and surface sanitization wipe passes.

- **D3: Domestic Tabletop Manipulation (Tasks 061–090)**: Common service robotics scenarios involving multi-colored block segregation, nested cup stacking, drawer articulation, and tabletop clutter clearing.
- **D4: Cleanroom Laboratory Sciences (Tasks 091–120)**: High-precision pharmaceutical and biological cleanroom tasks, including robotic pipette tip mounting, centrifuge tube placement, microplate well docking, and test tube rack transfers.
- **D5: Long-Horizon Multi-Stage Sequences (Tasks 121–150)**: Multi-phase manipulation tasks combining sequential tool acquisition, unlatching, re-orientation, and multi-part sub-assembly chaining.

### C. Physical Simulation and Sensing Modalities

All environments are simulated within native MuJoCo C++ physics with a standard control frequency of 20 Hz on a simulated 7-DoF Franka Emika Panda arm. The robot end-effector is equipped with a parallel-jaw gripper (0.0 m fully closed to 0.08 m fully open). Observations $\mathbf{o}_t \in \Omega$ captured at each timestep comprise:

- **Workcell Visual Stream**: $640 \times 660 \times 3$ uint8 RGB array providing an unoccluded, global third-person perspective of the workspace.
- **Wrist-Mounted Eye-in-Hand Stream**: $300 \times 420 \times 3$ uint8 RGB array rigidly attached to the gripper flange, capturing fine-grained object-gripper interaction.
- **Synchronized 3D Metric Depth**: Float32 depth maps $D \in \mathbb{R}^{H \times W}$ measured in true physical meters, enabling direct point-cloud back-projection via camera intrinsic matrix $\mathbf{K}$: $\mathbf{P} = \mathbf{K}^{-1}[u, v, 1]^T D(u, v)$.
- **Proprioceptive State**: 7-DoF joint positions $\mathbf{q} \in \mathbb{R}^7$, joint velocities $\dot{\mathbf{q}} \in \mathbb{R}^7$, end-effector Cartesian pose $\mathbf{x}_{\text{tcp}} \in \mathbb{R}^7$ (3D position and 4D unit quaternion), and continuous gripper opening width $w \in [0, 0.04]$ m.

## IV. THE AFFORDANCE-GROUNDED PERTURBATION MANIFOLD

To rigorously assess whether policy representations encode physical affordances or merely memorize visual landmarks, AFFORD-Bench defines a structured perturbation space $\mathcal{P} = \mathcal{P}_{\text{spatial}} \times \mathcal{P}_{\text{physics}} \times \mathcal{P}_{\text{visual}}$.

### A. Spatial-Kinematic Shifts ($\mathcal{P}_{\text{spatial}}$)

Spatial perturbations alter the initial 3D pose of the target manipulated object and receptacle:

$$\mathbf{p}_{\text{init}} \sim \mathcal{U}(\mathbf{p}_0 - \boldsymbol{\delta}_{\mathbf{p}}, \mathbf{p}_0 + \boldsymbol{\delta}_{\mathbf{p}}), \quad \theta_{\text{yaw}} \sim \mathcal{U}(-\theta_0, +\theta_0), \quad (2)$$

where $\boldsymbol{\delta}_{\mathbf{p}} = [\pm 0.06\,\text{m}, \pm 0.06\,\text{m}, 0.0\,\text{m}]^T$ and $\theta_0 = 30°$. Crucially, for insertion tasks, the radial clearance tolerance is systematically contracted from nominal $\epsilon_0 = 5.0$ mm down

to $\epsilon_{\text{tight}} = 1.5\,\text{mm}$, testing sub-millimeter visual servoing capabilities.

### B. Physical-Dynamic Shifts ($\mathcal{P}_{\text{physics}}$)

Unlike prior benchmarks that keep contact dynamics static, AFFORD-Bench randomizes surface Coulomb friction and inertial parameters:

$$\mu_{\text{contact}} \sim \mathcal{U}(0.10, 1.20), \quad m_{\text{obj}} \sim \mathcal{U}(0.5\,m_0, 3.0\,m_0), \quad (3)$$

where $\mu_0 = 0.70$ is the nominal friction coefficient and $m_0$ is the nominal mass. These variations directly test whether a learned policy can maintain grasp stability without slippage or excessive normal force generation.

### C. Visual-Photometric Shifts ($\mathcal{P}_{\text{visual}}$)

Visual perturbations preserve the physical geometry while modifying ambient and directional illumination intensity $I_{\text{light}} \in [0.4, 2.2]\,I_0$, light direction vector $\mathbf{v}_{\text{light}} \in \mathbb{S}^2$, and surface specular reflectance exponents.

### D. Composite OOD Shift ($\mathcal{P}_{\text{comp}}$)

The composite condition samples simultaneously across all three perturbation axes, generating compounded multi-source distribution shifts representative of uncontrolled real-world deployments.

## V. Algorithmic Oracles and Dataset Synthesis

### A. Deterministic Universal Kinematic Solvers

Collecting thousands of high-quality human demonstrations via teleoperation is prohibitively labor-intensive and frequently exhibits inconsistent sub-optimal behaviors [11]. AFFORD-Bench circumvents these bottlenecks by deploying deterministic, closed-loop algorithmic oracles constructed via analytical kinematic solvers and waypoint interpolation.

For each task family, the oracle solves for end-effector trajectories $\mathbf{x}_{\text{ee}}(t)$ using quintic minimum-jerk spline polynomials connecting pre-contact approach manifolds, contact alignment poses, insertion pathways, and release waypoints. Success is verified at every step using analytical kinematic predicates and simulation contact manifolds.

### B. Demonstration Scale and Packaging

Using unbuffered parallel workers, the synthesis pipeline generates **50 expert demonstrations per task** across all 150 tasks, yielding **7,500 total demonstrations** and over $1.5 \times 10^6$ **physical timesteps**. Generation runs locally at an effective throughput of $42.4\,\text{FPS}$ with zero external API calls or financial cost.

Each demonstration file is stored in standardized Robomimic HDF5 format:

$$\mathcal{D} = \{(\mathbf{o}_0, \mathbf{a}_0, r_0), (\mathbf{o}_1, \mathbf{a}_1, r_1), \ldots, (\mathbf{o}_{T-1}, \mathbf{a}_{T-1}, r_{T-1})\}. \quad (4)$$

Datasets include native train/validation mask splits (90% train, 10% validation) and full metadata attributes, enabling out-of-the-box training on existing VLA codebases including OpenVLA, SmolVLA, AtomVLA, Octo, and Diffusion Policy.

## VI. Training and Implementation Protocols

### A. Observation & Action Parameterization

The observation vector at timestep $t$ is $\mathbf{o}_t = (\mathbf{I}_{\text{work}}, \mathbf{I}_{\text{wrist}}, \mathbf{D}_{\text{3D}}, \mathbf{q}_t)$, where $\mathbf{I}_{\text{work}} \in \mathbb{R}^{224\times224\times3}$ is the work-cell view, $\mathbf{I}_{\text{wrist}} \in \mathbb{R}^{224\times224\times3}$ is the wrist camera view, $\mathbf{D}_{\text{3D}} \in \mathbb{R}^{224\times224\times1}$ is the aligned metric depth map, and $\mathbf{q}_t \in \mathbb{R}^{21}$ encapsulates joint angles, joint velocities, end-effector Cartesian pose, and gripper finger displacement. The policy action is parameterized as relative end-effector velocity $\mathbf{a}_t = [\Delta\mathbf{p}_t, \Delta\boldsymbol{\theta}_t, a_{\text{grip},t}] \in \mathbb{R}^7$, with spatial displacement $\Delta\mathbf{p}_t \in [-0.05, 0.05]\,\text{m}$, orientation change in axis-angle radians $\Delta\boldsymbol{\theta}_t$, and gripper clamp $a_{\text{grip},t} \in \{-1, +1\}$, normalized to $[-1, 1]$ across training demonstrations.

### B. Policy Architectures and Training Setup

We benchmark six representative policy architectures under uniform data regimes: (1) **Diffusion Policy (DP)** [4]: 1D Temporal U-Net with DDPM (100 steps, square-cosine schedule), action chunking horizon $H = 16$, $T_{\text{obs}} = 2$, trained via AdamW ($\text{lr} = 10^{-4}$) for 100k steps at batch size 64; (2) **Action Chunking with Transformers (ACT)** [20]: C-VAE with 8-layer Transformer predicting action chunks $H = 16$, trained using L1 + KL loss ($\beta = 10$) for 100k steps; (3) **OpenVLA (7B)** [9]: Llama-2-7B backbone with SigLIP vision encoder, fine-tuned via LoRA ($r = 32, \alpha = 64$) on attention projections with 256-bin action discretization for 30k steps per domain with AdamW ($\text{lr} = 5 \times 10^{-5}$); (4) **SmolVLA (450M)** [3]: Lightweight edge VLA based on SmolVLM-2 with a flow-matching action expert head, trained end-to-end for 50k steps; (5) **AtomVLA** [19]: Subtask-aware post-training incorporating LLM atomic decomposition and predictive latent world model rollouts; and (6) **3D Diffusion Policy (DP3)** [18]: Point-cloud-conditioned diffusion policy consuming $N_p = 2048$ downsampled points extracted from metric depth $\mathbf{D}_{\text{3D}}$.

## VII. Evaluation Protocol and Generalization Metrics

### A. Standardized Closed-Loop Rollout Procedure

Policy evaluation executes closed-loop at $20\,\text{Hz}$ interacting with the MuJoCo Franka Emika Panda physics environment ($500\,\text{Hz}$ solver, $dt = 0.002\,\text{s}$). For each task, evaluation is performed across $N = 20$ episodes initialized with pseudo-random seeds strictly disjoint from training demonstration seeds ($\mathcal{S}_{\text{test}} \cap \mathcal{S}_{\text{train}} = \emptyset$). Maximum episode horizon is set to $H_{\max} = 400$ steps for single-stage manipulation (D1–D4) and $H_{\max} = 800$ steps for long-horizon sequences (D5).

### B. Standardized Evaluation Suites

All models are evaluated across five rigorously defined suites: (1) **Nominal (In-Distribution)**: Default scene layout, initial poses $\mathbf{p}_0 \pm \boldsymbol{\delta}_{\text{small}}$ ($< 1\,\text{cm}$), nominal friction $\mu_0 = 0.70$, nominal mass $m_0$, and standard illumination $I_0$; (2) **Spatial Generalization** ($\mathcal{P}_{\text{spatial}}$): Uniform translation offsets $\pm 6\,\text{cm}$, yaw rotations $\pm 30°$, and contracted insertion tolerances $\epsilon \leq 1.5\,\text{mm}$; (3) **Physical Generalization**

TABLE II
**BENCHMARK SUCCESS RATES (%) ACROSS DOMAINS.** EVALUATED OVER 20 TEST ROLLOUTS PER TASK UNDER NOMINAL AND COMPOSITE OOD ($\mathcal{P}_{\text{COMP}}$) PERTURBATION CONDITIONS.

| Task Domain | Diffusion Pol. | | OpenVLA-7B | | SmolVLA-450M | |
|---|---|---|---|---|---|---|
| | Nom. | OOD | Nom. | OOD | Nom. | OOD |
| D1: Industrial Precision | 86.4% | 14.2% | 81.0% | 11.5% | 74.2% | 8.5% |
| D2: Culinary Operations | 91.2% | 23.0% | 88.5% | 18.2% | 82.0% | 14.0% |
| D3: Domestic Tabletop | 94.0% | 31.5% | 92.0% | 27.0% | 87.5% | 21.0% |
| D4: Laboratory Sciences | 83.5% | 9.8% | 78.4% | 8.0% | 71.0% | 6.2% |
| D5: Long-Horizon Sequences | 72.0% | 6.5% | 67.2% | 4.2% | 59.5% | 3.0% |
| **Aggregate Average** | **85.4%** | **17.0%** | **81.4%** | **13.8%** | **74.8%** | **10.5%** |

($\mathcal{P}_{\text{physics}}$): Friction variation $\mu \in [0.10, 1.20]$ and mass scalings $m \in [0.5, 3.0]\, m_0$, testing non-slip grasp stability and force compliance; (4) **Visual Generalization** ($\mathcal{P}_{\text{visual}}$): Illumination intensities $I \in [0.4, 2.2]\, I_0$, random directional light vectors, and camera pose jitter ($\pm 2^\circ$); and (5) **Composite OOD** ($\mathcal{P}_{\text{comp}}$): Simultaneous joint perturbation across all three axes.

### C. Quantitative Evaluation Metrics

We measure policy competence using four formal metrics: (i) **Task Success Rate (SR)**: Binary terminal affordance satisfaction $\text{SR} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}(\mathbf{s}_T^{(i)} \in \mathcal{C}_{\text{aff}})$; (ii) **Physical Robustness Index (PRI)**: Fraction of competence preserved under perturbation $\text{PRI}(\mathcal{P}) = (\text{SR}(\mathcal{P})/\text{SR}(\text{Nominal})) \times 100\%$; (iii) **Subgoal Completion Ratio (SCR)**: For D5 multi-stage sequences, tracking fractional milestone achievement $\text{SCR} = \frac{1}{NM}\sum_{i=1}^{N}\sum_{m=1}^{M} \mathbb{I}(\text{Subgoal } m \text{ reached})$; and (iv) **Force Violation Rate (FVR)**: Percentage of episodes terminating prematurely due to excessive contact reaction forces ($F_N > 50\,\text{N}$) or kinematic jamming.

## VIII. EMPIRICAL RESULTS, HYPOTHESES & DIAGNOSTICS

### A. Core Scientific Hypotheses

We formalize three fundamental scientific hypotheses that AFFORD-Bench is designed to test:

- **Hypothesis 1 ($H_1$: Affordance Collapse vs. Appearance Memorization)**: *Policies that achieve $> 85\%$ nominal success by memorizing 2D visual trajectories will suffer asymmetric catastrophic failure ($> 70\%$ relative drop) under physical-dynamic shifts ($\mathcal{P}_{\text{physics}}$), even when the spatial coordinates of target objects are identical to training.*
- **Hypothesis 2 ($H_2$: Precision-Tolerance Fragility)**: *The performance degradation curve under spatial perturbation is non-linear and governed by kinematic clearance $\epsilon_{tol}$. Tasks in D1 and D4 (clearance $\leq 3\,mm$) will experience collapse at spatial perturbations an order of magnitude smaller than domestic tasks in D3.*
- **Hypothesis 3 ($H_3$: Invariance via 3D Metric Perception)**: *Policies that incorporate aligned 3D metric depth representations (e.g., DP3) will exhibit significantly higher resistance to visual-photometric shifts ($\mathcal{P}_{\text{visual}}$) compared to 2D-RGB policies.*

### B. Empirical Insights and Failure Diagnostics

Table II presents preliminary evaluation results across 20 evaluation episodes per task. Under nominal conditions, Diffusion Policy, OpenVLA, and SmolVLA exhibit strong task completion across all five domains ($85.4\%$, $81.4\%$, and $74.8\%$ aggregate success, respectively).

However, under composite out-of-distribution shifts ($\mathcal{P}_{\text{comp}}$), performance catastrophically degrades across all architectures down to $17.0\%$, $13.8\%$, and $10.5\%$. Diagnostic analysis reveals distinct failure signatures:

1) **Token-Compression Bottlenecks in Compact VLAs**: While SmolVLA provides outstanding throughput ($> 50\,\text{Hz}$), its visual token reduction (64 tokens) hinders fine-grained spatial servoing, leading to higher failure rates in sub-centimeter insertions in D1 and D4.
2) **Grasp Slippage vs. Subtask Planning**: In long-horizon tasks (D5), AtomVLA's predictive world model effectively prevents subtask sequencing errors; however, when surface friction $\mu \leq 0.25$, open-loop gripper commands still fail during lifting phases, showing that high-level planning cannot compensate for low-level contact non-compliance.
3) **Kinematic Jamming**: In D1 and D4, minor rotational misalignments during insertion phases produce high reaction forces, jamming the manipulator due to rigid impedance gains.

## IX. DISCUSSION AND STAGE 2 EXTENSION CHARTER ROADMAP

In accordance with the RSS 2027 Shepherding Process, this Stage 1 Extended Abstract establishes the foundational methodology, task taxonomy, qualitative rendering suite, training/evaluation protocols, and affordance perturbation manifold of AFFORD-Bench. For the Stage 2 final submission, we outline our planned Extension Charter:

- **Full Empirical Scaling Across All 150 Tasks**: Executing complete evaluation matrices (50 rollouts per condition $\times$ 150 tasks) across Diffusion Policy, ACT, OpenVLA, SmolVLA, AtomVLA, and DP3.
- **Disentangled Perturbation Ablations**: Isolating the independent effects of $\mathcal{P}_{\text{spatial}}$, $\mathcal{P}_{\text{physics}}$, and $\mathcal{P}_{\text{visual}}$ to map individual sensitivity frontiers.
- **Physical-to-Real Sim-to-Real Pilot**: Evaluating policy transfer onto a physical Franka Emika Panda workcell for representative D1 (gear meshing) and D4 (pipette docking) tasks.

By uniting high physical precision, diverse real-world domains, rigorous protocols, and affordance-grounded perturbations, AFFORD-Bench provides the robotics community with an uncompromising benchmark to steer policy learning toward genuine physical generalizability.

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